

David Marr

Creative Systems Engineer — AI Platforms, Infrastructure & Prototyping

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Summary

Systems engineer who builds the substrate underneath creative work — APIs, services, pipelines, and platform primitives that let designers, creative technologists, and product people prototype and ship faster.

Twenty-five years wiring production systems together on Python, TypeScript, Node.js, Docker, Kubernetes, and AWS. I treat agentic AI as a new programming primitive, run a home Kubernetes cluster to learn by doing, and am most at home in ambiguous, experiment-driven environments.

Experience

Context Labs, Principal Software Engineer / Engineering Lead (Platform) – Cambridge, MA June 2022 – Mar 2026

Built platform infrastructure for a multi-tenant analytics product — the APIs, services, and integration layer that product and visualization teams built on top of.

- Built production integration layer in Node.js/TypeScript (GraphQL BFF) wiring Java APIs, databases, auth, and localization into a single surface product teams could move on quickly.
- Drove adoption of AI coding assistants and MCP tooling across the stack — hands-on workshops, review guardrails, and platform patterns so teams could experiment with agents without breaking production.
- Shaped shared platform primitives (data access, charting with Vega Lite, geospatial with Mapbox) that reduced repeated plumbing work for feature teams.
- Led distributed engineering across three continents; introduced lightweight delivery practices that improved velocity and cut incident recovery time by 50%.
- Mentored engineers on systems thinking, service boundaries, and how to prototype safely against real infrastructure.

Skills used: Python, Node.js, TypeScript, GraphQL, PostgreSQL, Docker, Kubernetes, Vue, Nuxt

Filtered.ai, Lead Engineer – Boston, MA

Feb 2020 – Feb 2022

Lead engineer wiring models, services, and data pipelines into a hiring-assessment product on AWS and PostgreSQL.

- Integrated computer-vision models and human-in-the-loop validation (AWS Mechanical Turk) into scoring pipelines that had to be reliable in production.
- Built ingestion, scheduling, grading, and workflow services that connected product requirements to running infrastructure.
- Partnered with data science on model deployment, quality checks, and operational monitoring for automated assessment at scale.

Skills used: Python, AWS, PostgreSQL, Node.js, TypeScript

Turo, Staff Software Engineer – Cambridge, MA

Apr 2018 – Dec 2019

Owned backend systems for trip management and dynamic pricing on PostgreSQL — the machinery behind product-facing availability and rate logic.

- Maintained pricing and host-recognition systems where product rules depended on dependable service and data plumbing.
- Worked with product and operations to translate fuzzy requirements into shippable backend behavior.

Skills used: PostgreSQL, TypeScript, JavaScript, React

OneMarket / Westfield Labs, Senior Software Architect – San Francisco, CA

Jan 2016 – May 2018

Wired together hardware, payments, and web services into a smart parking platform — facial recognition, billing, gates, and venue software talking through APIs and databases.

- Integrated disparate systems — cameras, payment rails, gate hardware, and web apps — into a product that generated ~\$2.7m in monthly recurring revenue.
- Built indoor mapping and kiosk software across 4 flagship shopping centers, averaging 60,000+ monthly sessions.

Skills used: Node.js, PostgreSQL, C, JavaScript, Stripe, Mapbox

Tilt.com, Senior Systems Engineer – San Francisco, CA

July 2013 – Dec 2015

- Pioneered server-rendered React integrated with a Perl/Dancer backend — early work bridging new UI patterns to existing services.
- Implemented and iterated a simplified contribution flow yielding 15% lift in site signups.

Skills used: Perl, JavaScript, PostgreSQL, React

Various, Previous Leadership & Technical Roles – Massachusetts & California

2000-2013

- Design-to-production prototyping at Frog Design — translating creative concepts into working software under tight ambiguity.
- Co-founded early social product (Secret Feature); publishing and ad-tech platform work at Say Media.
- Scaled frontend architecture for high-traffic Yahoo properties; enterprise integrations and industrial automation dashboards.

Skills used: Node.js, JavaScript, MySQL, PHP, .NET, HTML, CSS

Systems

Personal homelab (Kubernetes / GitOps)

Self-hosted infrastructure lab documented on marr.github.io — Talos Linux, Flux, SOPS secrets, and service orchestration on a home cluster.

- Operate and evolve a Kubernetes cluster to understand containers, networking, storage, and GitOps reconciliation in practice.
- Built custom observability dashboards and agent-oriented runbooks so debugging stays fast as the environment grows.

Education

University of New Hampshire, BS in Business Administration – Durham, NH

Sept 1996 – May 2000

Focus on Management of Information Systems (MIS)

Skills

Systems & Infrastructure: Docker, Kubernetes, AWS, Python, Node.js, TypeScript, PostgreSQL, GraphQL, Bash, Git, MongoDB

AI, Agents & Prototyping: Vercel AI SDK, MCP, Cursor, Opencode, Agent Skills, Cloudflare Workers, Cloudflare Durable Objects, Pi

Web & Visualization: Vue, Nuxt, React, Vega Lite, Mapbox, D3, WebGL, WebRTC, Go